

iProov Presentation Screen Attack Detection

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1 Objective and data

1.1 Task and proposed solution

The objective was to build a presentation attack detection (PAD) prototype that takes face images and produces a continuous attack score for each image. A successful PAD prototype will be able to distinguish bona fide presentations, where a real person is in front of the camera, from screen attacks, where a camera captures a face displayed on another device. The unique challenge of this problem is the limited data sample, with only 23 available screen attack samples and 1,000 bona fide.

After various experiments, my final solution is a PatchCore-inspired, one-class anomaly detector combining two views of each image. A whole-image branch examines the complete frame after resizing, while a native-crop branch examines nine central crops without resizing their pixels. Both use a frozen ImageNet-pretrained ResNet-18 to extract local features and compare them with separate bona fide reference banks. Their scores are normalised using bona fide calibration data and combined through a maximum rule.

The final model achieved an ROC AUC of 0.9493. At the selected threshold, it detected 13 of 23 attacks and rejected 2 of 150 bona fide images. These are promising results that supports further research into the combined approach, but it is important to note the 43.5% screen attack acceptance rate as well as other limitation that will be later discussed.

1.2 Dataset and initial observations

The supplied dataset contained 1,000 bona fide images and 23 screen attacks. From my initial observations, there are several data characteristics that influenced both my experiment design and evaluation choices:

- **Limited attack coverage.** 23 attacks provide little evidence about the diversity of displays, cameras and conditions that a deployed system could encounter. Splitting these examples between supervised training and evaluation would further reduce the evidence available for each purpose.
- **Matching source pairs.** Each attack had a corresponding bona fide image depicting the same source content, shown in Figure 1. These pairs created a leakage risk if the bona fide source entered reference fitting while its recaptured attack was used for evaluation. I therefore reserved both members of every known pair for evaluation.
- **Resolution and aspect-ratio differences.** All bona fide images were $1,024 \times 1,024$ pixels, while attacks had larger, variable dimensions and were not consistently square. In fact, a simple rules based classifier such as $(\text{image_width} > 1024)$, would perfectly separate the two classes. This is a possible confounder, as a model could learn to differentiate from resolution rather than learning from screen-attack signals.

- **Variation in visible artefacts.** Some attacks contained clear device borders or user-interface elements. Others showed finer repetitive texture, colour distortion or blur. This suggested that a single image scale might not preserve all useful information.
- **Variation in Bona Fide Images.** While all bona fide images have standardised resolutions and relative positions of facial features, many images have unique and distinct features. For example, there is a large variety of accessories, clothes, hats, backgrounds and etc, that can make any image sample distinctive. This can influence models focused on anomaly detection.

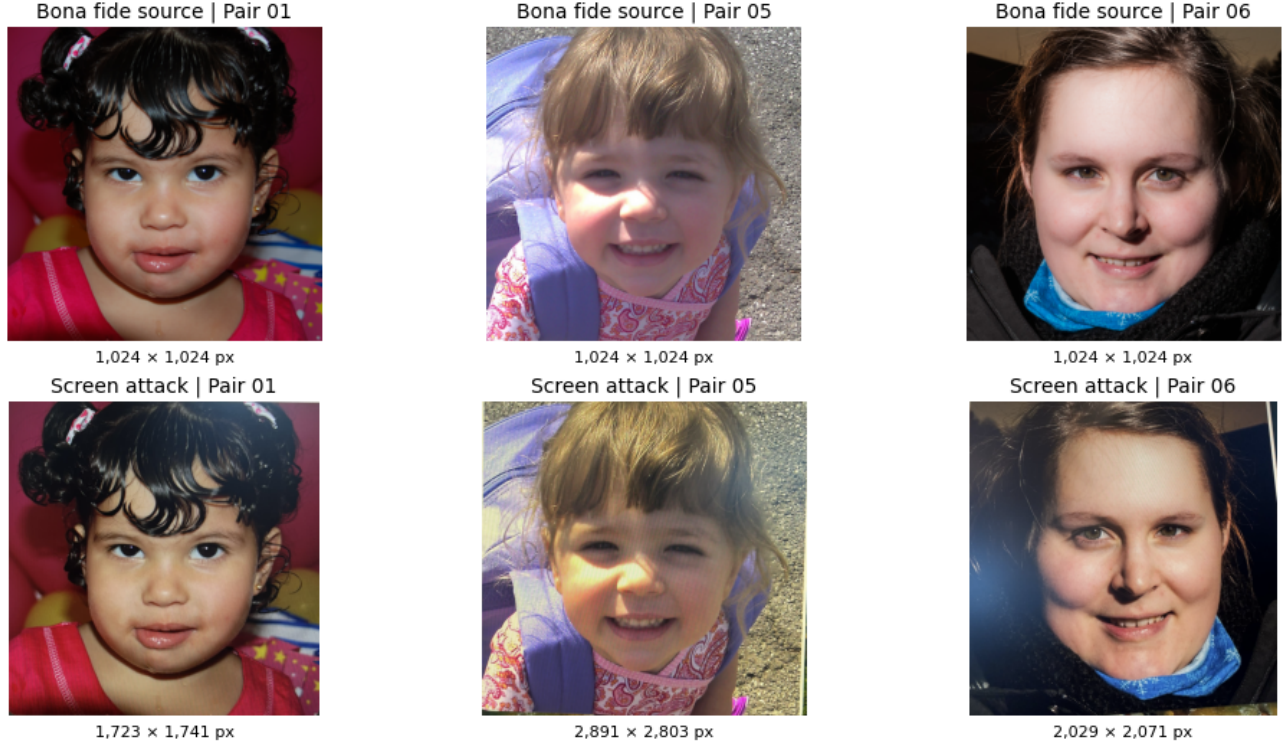


Figure 1: Sample Bona Fide and Presentation Attack Pairs

2 Methodology

2.1 Whole-image anomaly detection

Motivation: The limited number of screen attacks was what led me to explore one-class anomaly detection models. Training a supervised classifier with a sample size of 23 real screen attacks is not feasible, and although I initially explored creating synthetic data (discussed in section 5) to counteract this, the large distribution of screen attacks made this non-trivial. Using anomaly detection to model the bona fide reference distribution instead, was a simpler baseline approach that did not require modelling the full distribution of screen recaptures in screen attacks.

There are many different anomaly detection models, but I chose experiment with PatchCore as a starting point for various reasons. What was most promising about PatchCore was that it extracts intermediate layers of a frozen ResNet model to build a memory bank of features. These features are where the textures and edges of an image are preserved, which is what is often used to identify screen attack images through moiré patterns and texture distortions. PatchCore therefore offered a practical way to identify screen attacks without fine-tuning the network on the limited PAD dataset. The resulting baseline adapted PatchCore’s local-feature memory-bank approach to PAD. It comprised three stages: extracting local image descriptors, constructing a bona fide reference bank, and scoring new images by their distance from that reference.

Approach: Each full image was resized to 256×256 pixels and normalised using ImageNet channel statistics. Features were extracted from ResNet-18’s `layer2` and `layer3`, combining information from two levels of the network. A 3×3 average-pooling operation aggregated neighbouring features within each feature map. The `layer3` map was then bilinearly interpolated to match the spatial dimensions of `layer2`. The feature vectors from each layer were L2-normalised, concatenated at corresponding spatial locations, and L2-normalised again. This produced a 32×32 grid of 384-dimensional descriptors, giving 1,024 local descriptors per image. Each descriptor represented a region of the input through the network’s receptive field, rather than a separately extracted image crop. These regions overlapped and could capture both local appearance and surrounding context.

The same feature-extraction procedure was applied to the 650 bona fide reference images. To limit storage and nearest-neighbour computation, 128 candidate descriptors were sampled from each image, and the candidate pool was reduced to a bank of 10,000 descriptors using seeded random-priority sampling. This bank represented examples of local appearance in the bona fide reference set.

The bank retained individual descriptors rather than reducing all bona fide images to a single average representation. A query region could therefore be compared with its closest available bona fide example, accommodating variation within the reference data.

For a new image x , each local descriptor $q_i(x)$ was compared with every descriptor in the reference bank $\mathcal{M}$. Its local anomaly score was the Euclidean distance to its nearest neighbour:

$$d_i(x) = \min_{m \in \mathcal{M}} \|q_i(x) - m\|_2.$$

A larger distance indicated that the region’s feature representation was less similar to the stored bona fide examples. The image-level score was the mean of the largest 10% of local distances. Writing the distances in descending order as $d_{(1)}(x) \geq \dots \geq d_{(N)}(x)$:

$$s_{\text{whole}}(x) = \frac{1}{k} \sum_{j=1}^k d_{(j)}(x), \quad k = \lceil 0.10N \rceil.$$

With $N = 1,024$ descriptors, this averaged the 103 largest distances. A decision threshold was subsequently calibrated using separate bona fide images.

2.2 Native-resolution crop anomaly detection

Motivation: Inspecting the whole-image baseline’s errors motivated me to take a closer look at local texture. Some successful detections showed patterns of clear device borders or interface elements, whereas some missed attacks contained repetitive patterns and texture differences that were more visible when examining the zoomed-in original images (Shown in Figure 5). Combined with the high resolution of screen attack images, this suggested that resizing the full image could weaken useful screen-recapture cues, including moiré-like patterns. I therefore experimented with a crop-based approach that preserved native pixel detail.

Approach: The crop detector extracted a central 384×384 -pixel region and divided it into nine contiguous, non-overlapping 128×128 -pixel crops, as shown in Figure . Each crop was processed by the same frozen feature extractor without image or crop resizing. Crop locations were deterministic rather than randomly resampled at inference. The detector used a separate bona fide reference bank because native crops and resized whole images produce different feature distributions. As in the whole-image baseline, local features were compared with their nearest reference features. Distances from all nine crops were pooled, and the largest 10% were averaged to obtain a single image score. This was therefore not an average of nine independently computed crop scores.

The crop detector produced a different pattern of detections and errors from the whole-image baseline, motivating investigation of their combined use. However, fixed-size native crops also covered different proportions of faces when image resolutions differed. Their performance could therefore reflect both useful texture information and the dataset’s resolution confound.

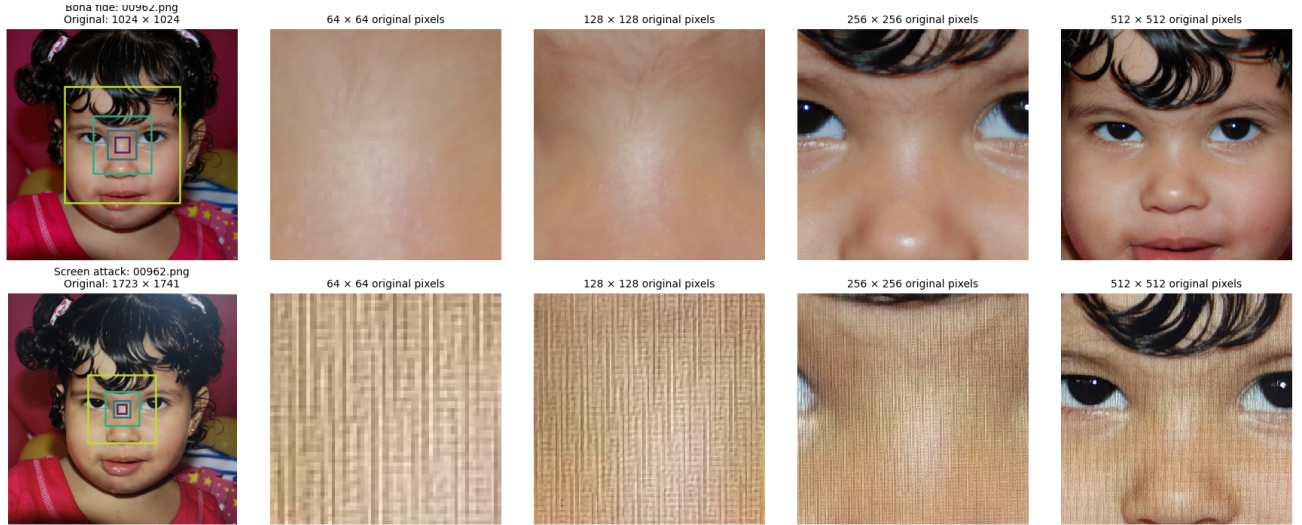


Figure 2: Inspection of local image detail in bona fide images and screen attacks, motivating the native-resolution crop approach.

2.3 Fusion of whole-image and crop scores

Motivation: After experimenting with both previously discussed approaches, both approaches showed useful but incomplete discrimination, with different patterns of successful detections and missed attacks. Qualitative inspection of the results suggested that whole-image processing could retain useful broad

context such as the physical device of screen attacks, while native crops could preserve local textures from the screen artefacts. This complementary behaviour led me to combine the approaches to create a fusion model, so that strong evidence from either the whole-image and native-crop detectors can to the final classification.

Approach: The combined system uses two parallel processing paths, referred to as the whole-image and native-crop branches. Each branch produces a raw anomaly score using its own bona fide reference bank. To align their score scales, I estimated the median and interquartile range of each branch’s scores using a separate bona fide normalisation set. For branch b , the normalised score was:

$$z_b(x) = \frac{s_b(x) - m_b}{\text{IQR}_b},$$

where $s_b(x)$ is the raw anomaly score, and m_b and IQR_b are the corresponding bona fide median and interquartile range.

The final attack score was the maximum of the two normalised scores:

$$S(x) = \max(z_{\text{whole}}(x), z_{\text{native}}(x)).$$

The maximum rule retained a strong anomaly signal from either branch without averaging it down when the other branch scored the image as more typical. No fusion weights were learnt from attack examples. A new threshold was calibrated on the combined score using a separate bona fide threshold-calibration set, accounting for the joint behaviour of both branches. The final output is an anomaly score rather than an attack probability: values can be negative or greater than one.

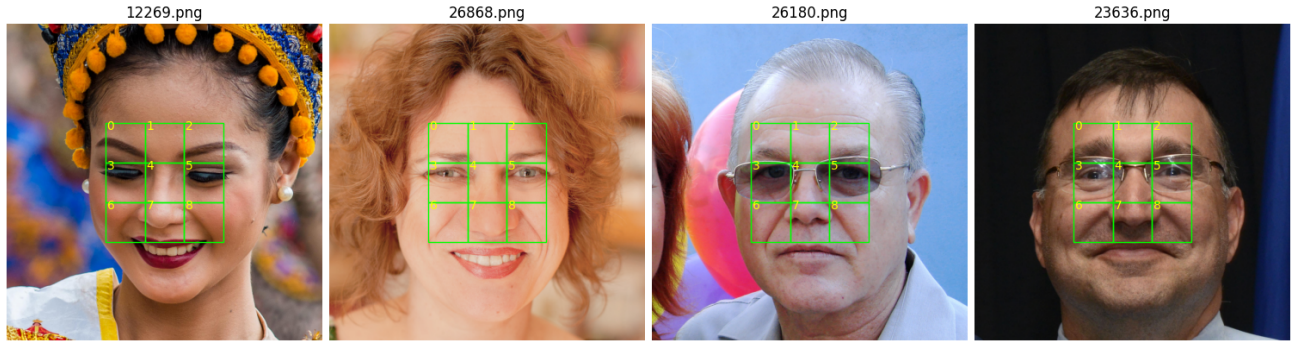


Figure 3: Example of Crop Extraction Process

3 Experimental setup and results

3.1 Data separation and calibration

The final experiment used the following data split:

Purpose	Bona fide images	Screen attacks
Reference bank construction	650	0
Score normalisation	100	0
Threshold calibration	100	0
Exploratory evaluation	150	23

The original 200-image bona fide calibration set was divided into two groups of 100 by sorting source identifiers and shuffling with seed 42. Both branches used the same subsets. All 23 attack-source counterparts were included in the bona fide evaluation set and excluded from reference fitting and calibration.

Case-normalised filenames were used to identify source pairs. The fusion workflow checked that the two branches used matching source groups, partitions and pixels, and that their checkpoints contained identical frozen backbone weights. These controls addressed direct pair leakage, but did not establish complete separation of identities or capture sessions.

The final threshold targeted a nominal 5% bona fide presentation classification error rate (BPCER). For n bona fide calibration scores, I selected ascending rank $\text{ceil}((n + 1) \times 0.95)$. With 100 images, this was the 96th smallest score. An image was classified as an attack only when its score was strictly greater than the threshold. With distinct calibration scores, this rejects four of the 100 images; the target does not guarantee 5% rejection on new data.

The saved fusion threshold was approximately 1.678816. The threshold were fixed before the final evaluation pass. However, the attack images had already influenced the broader development process, so this evaluation is leaning towards exploratory rather than an pure untouched test of a prespecified method.

3.2 Implementation settings

Component	Setting
Backbone	Frozen ImageNet-pretrained ResNet-18 in evaluation mode
Feature layers	layer2 and layer3
Feature processing	3×3 average pooling; bilinear alignment of layer3 to layer2; per-layer L2 normalisation; concatenation and final L2 normalisation
Local descriptor	384 dimensions
Whole-image input	Full frame resized to 256×256
Native input	Nine central 128×128 crops without resizing
Reference candidates	128 per whole image; 14 per native crop, or 126 per image
Memory bank	10,000 descriptors per branch, selected through seeded random-priority sampling
Local distance	Exact nearest-neighbour Euclidean distance to the corresponding bank
Image aggregation	Mean of the largest 10% of local distances

Component	Setting
Fusion	Maximum of median/IQR-normalised branch scores

The whole-image branch produced 1,024 descriptors and averaged its 103 largest distances. The native branch produced 2,304 descriptors across nine crops and averaged its 231 largest distances. Exact nearest-neighbour distances were calculated in chunks to control temporary memory use.

The implementation is PatchCore-inspired. It uses random-priority bank reduction in place of greedy coresets selection and its own top-fraction score aggregation. The native branch preserves input pixels, but the backbone still performs convolutional downsampling internally.

3.3 Evaluation metrics and primary comparison

I reported ROC AUC to assess ranking across thresholds, alongside threshold-specific error rates and raw counts. APCER is the proportion of attacks accepted as bona fide; BPCER is the proportion of bona fide images rejected as attacks. Reporting both avoids hiding security failures behind the larger bona fide class.

For the primary comparison, both standalone branches were recalibrated using the same 100-image threshold subset and nominal 5% target as fusion. The observed evaluation BPCER was allowed to differ.

Model	ROC AUC	Attacks detected	Detection rate	APCER	Bona fide rejected	BPCER
Whole image	0.7974	10/23	43.48%	56.52%	13/150	8.67%
Native crops	0.8513	10/23	43.48%	56.52%	1/150	0.67%
Maximum fusion	0.9493	13/23	56.52%	43.48%	2/150	1.33%

Fusion achieved the highest observed AUC and detected three more attacks in aggregate than either standalone branch. Compared with the native branch, this came with one additional bona fide rejection. Compared with the whole-image branch, it produced both more attack detections and fewer bona fide rejections. These are observed differences on this small population, not evidence of statistically established superiority across capture conditions.

Using a simple binary OR classification rule of the whole-image and native crop approach would results in 17/23 attacks but rejected 14/150 bona fide images. This used the original thresholds of each model and was not the calibrated maximum-fusion system. Its higher detection count therefore should not be substituted for the final model’s 13/23 result.

3.4 Qualitative Results

Bellow are examples of the highest and lowest score samples from the whole-image and native-crop detectors

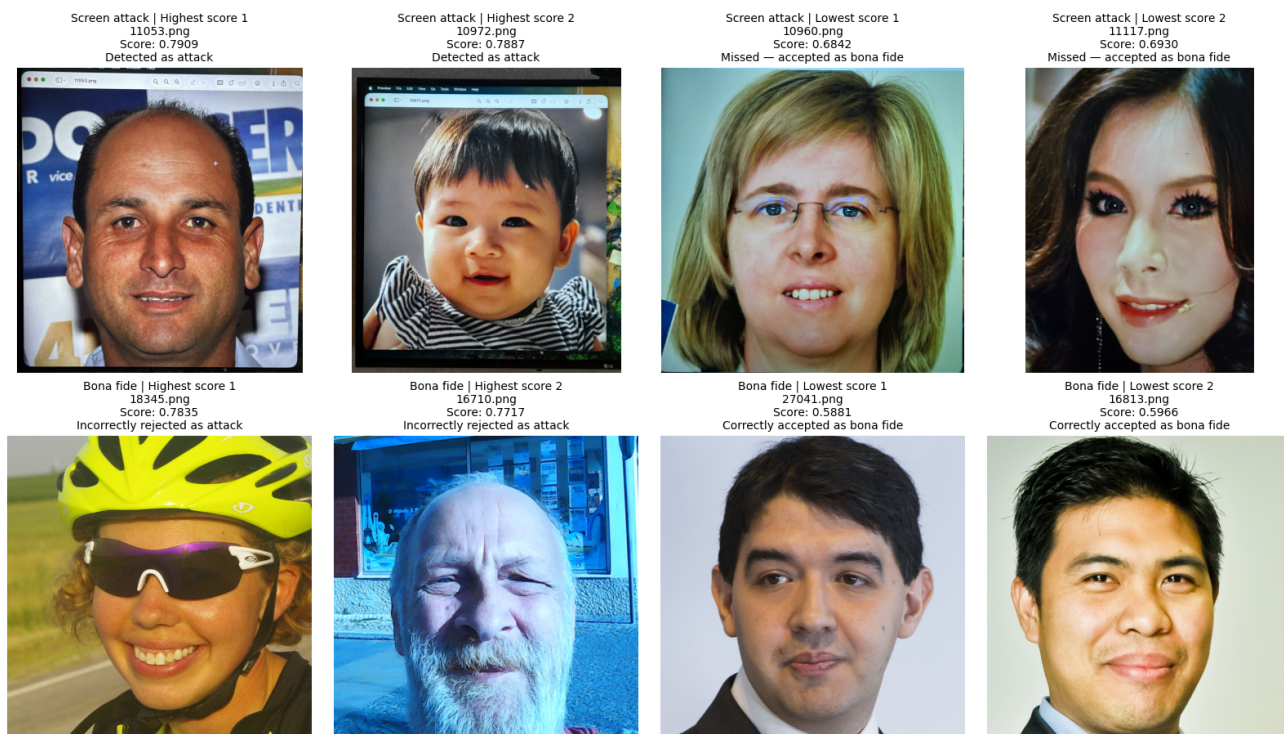


Figure 4: Highest and lowest scoring images for Local Image Detector

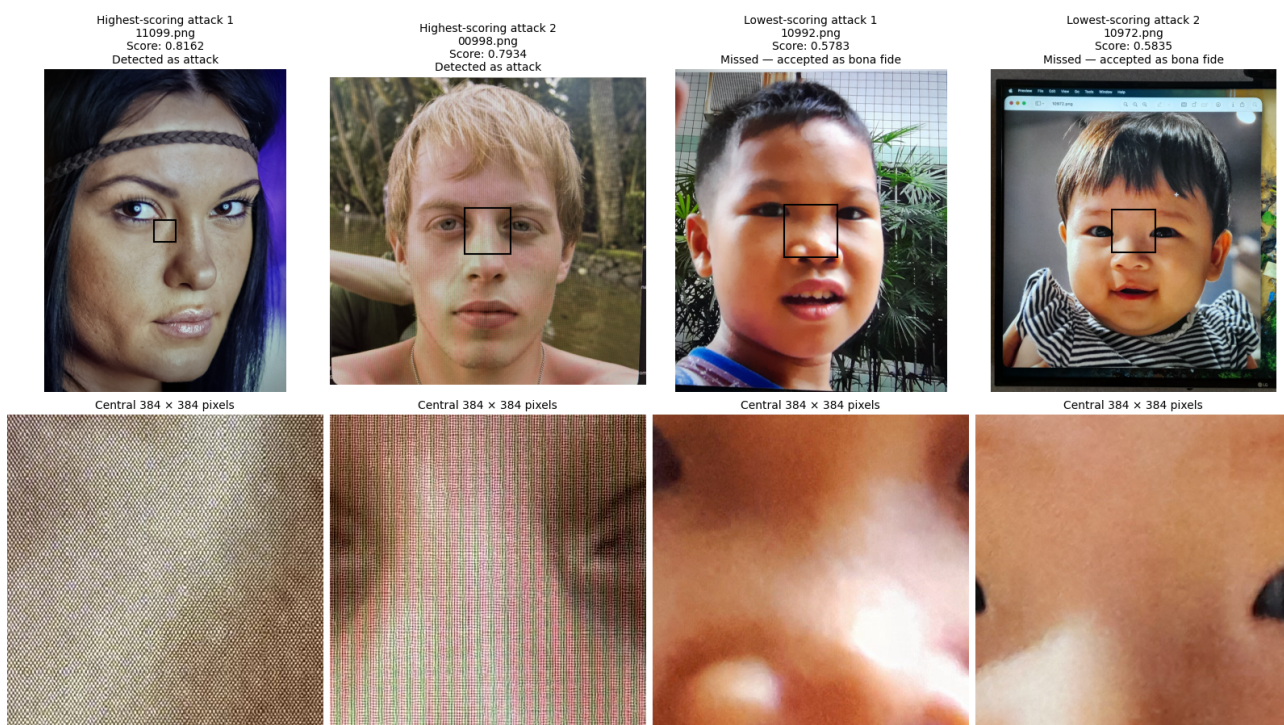


Figure 5: Highest and lowest scoring images for Native-Crop Detector

4 Analysis and discussion

4.1 Interpretation of the results

The results support the hypothesis that whole-image and native-crop representations provide complementary information. Reviewing the qualitative results for the the whole-image detector, we can observe that the highest assigned attack score are examples of screen attacks containing clear screen borders and UI elements. It struggled however, with images lacking these contextual clues and often incorrectly rejects images with atypical feature (such as the sunglasses in Figure. On the other hand, The native-crop detector’s highest-scoring attacks contained pronounced repetitive patterns in their central regions, whereas its lowest-scoring attacks showed smoother texture with less obvious screen structure. A particularly informative example is 10972.png: it was among the highest-scoring attacks for the whole-image detector but among the lowest for the native-crop detector. Its visible screen border and interface lie outside the central crop, illustrating how the two approaches retain different information.

Fusion improved ranking and increased attack detection relative to both recalibrated branches, while retaining a low observed bona fide rejection rate. The gain is practically interesting because it did not require fitting network parameters or fusion weights to the 23 attacks.

Nevertheless, a ROC AUC of 0.9493 does not mean that 94.93% of attacks were detected. AUC describes relative score ordering across thresholds. At the submitted operating point, ten of 23 attacks were accepted. Lowering the threshold could increase detection, but would also risk rejecting more bona fide users. The current result demonstrates useful separation with a substantial remaining security limitation.

4.2 Evidence supporting the result and its limits

Several controls increase confidence that the implementation measures the intended experiment. Known source pairs were separated from fitting and calibration; duplicate checks reduced accidental overlap; both branches used the same populations; and score scaling and threshold selection used bona fide data only. Per-image scores and branch diagnostics made it possible to inspect which decisions contributed to aggregate metrics.

The deployment package also separates model fitting from inference. The export workflow specifies comparison against all 173 notebook evaluation rows, checking raw scores, normalised scores, fusion scores and decisions. The recorded Docker smoke run processed 23 images offline without reported errors, although complete 173-image Docker parity remained pending. These checks support software consistency and should not be interpreted as independent evidence of PAD generalisation.

The strongest limitation is development-time selection bias. I repeatedly inspected the same attacks while deciding which approaches to pursue. Although their scores were not used to fit the final calibration parameters, they influenced model design. The observed performance may therefore be optimistic relative to a genuinely unseen attack set.

Statistical precision is also limited. One attack changes the measured detection rate by approximately 4.35 percentage points. Correlated captures and shared acquisition characteristics can reduce the effective sample size further. I therefore trust these results as evidence of a functioning prototype and useful experimental direction, but not as a reliable estimate of deployment performance.

4.3 What the method may be responding to

The model measures feature-space deviation from bona fide reference images. It has no explicit representation of whether a person is live, whether a screen is present, or whether a texture is moiré.

Visual inspection suggests that broad structural cues may contribute to the whole-image scores, while local recapture texture may contribute to native-crop scores. However, the two branches could also respond to differences in sharpness, colour processing, image scale or background. The branch supplying the maximum score identifies which representation dominated a decision, not the visual cause of that decision.

Resolution is particularly important. Every bona fide image was $1,024 \times 1,024$, whereas attacks were larger. A fixed crop consequently covered different fractions of the image and potentially different fractions of a face. Whole-image resizing also applied different amounts of resampling across classes. Even without explicitly feeding dimensions into the model, those preprocessing consequences can remain visible in its features.

Controlled interventions are needed to distinguish these explanations. For example, removing visible borders, applying comparable compression to both classes, or controlling facial scale could reveal how much performance depends on each cue. Such interventions must be interpreted carefully because resizing and filtering can themselves remove genuine screen artefacts.

4.4 Expected failure modes

Condition	Why performance may degrade
Sharp, border-free recaptures	Attacks may lack conspicuous structural or texture deviations from the reference images
Blur, compression or prior downsampling	Fine screen texture may be weakened before the native branch receives the image
Different cameras or processing pipelines	Bona fide images may appear anomalous because their capture characteristics differ from the reference bank
Off-centre faces or altered framing	Central crops may miss useful facial regions or sample mostly background
New image resolutions or facial scales	Fixed pixel crops may represent a different visual scale from the reference data
Unusual bona fide lighting or backgrounds	General visual novelty may be mistaken for attack evidence

Maximum fusion also allows an unreliable branch to dominate when its normalised score is high. Calibration controls the observed bona fide distribution, but cannot guarantee behaviour under distribution shift. The prototype is restricted to single images and does not exploit temporal or active-challenge information.

5 Other experiments and future work

5.1 Synthetic attack training

An earlier experiment fine-tuned a ResNet-18 binary classifier using bona fide images and procedurally generated screen-like augmentations. The aim was to address the scarcity of real attacks by simulating recapture characteristics, including moiré-like patterns. Real screen attacks were reserved for evaluation.

The classifier achieved ROC AUC 0.9876 on synthetic evaluation data and detected 140/150 synthetic attacks. On the real attacks, AUC fell to 0.6780 and only 8/23 were detected. In both reported evaluations, 11/150 bona fide images were rejected. These results came from the earlier experiment’s own protocol and should not be treated as a controlled ablation of the final fusion method.

The large synthetic-to-real gap suggested that the simulated artefacts did not adequately represent the real recapture distribution. It may also reflect other differences between the evaluation populations, so it does not isolate augmentation realism as the only cause. The experiment nevertheless showed that strong synthetic performance was insufficient evidence of real-attack detection and helped motivate the one-class approach.

5.3 Priorities with more time and data

Independent evaluation would be the first priority. I would freeze the current system and evaluate it on new attacks and bona fide captures spanning additional displays, cameras, people, sessions and lighting conditions. Development and final evaluation should be separated at the relevant source, identity and acquisition levels where metadata permits. This would test whether the observed complementarity survives beyond the examples that shaped the method.

The next priority would be controlling acquisition confounds. I would collect genuine and attack images through comparable capture and processing pipelines, with overlapping resolutions and facial scales. I would also run paired preprocessing interventions to measure sensitivity to borders, compression, blur and scale, recording both attack misses and bona fide rejections.

Model changes should then follow the failure analysis. Useful ablations include face-relative or multi-scale crops, different local-distance aggregation fractions, and more representative memory-bank reduction. I would compare each change against the fixed baseline on a separate development set. Increasing crop coverage may capture more evidence, but also increases computation and exposure to irrelevant background, so it should be evaluated as a trade-off.

Calibration and deployment behaviour need broader validation. More bona fide calibration data would support more stable estimation of low-rejection operating points. I would assess attack detection across an agreed range of BPCER values, evaluate important capture subgroups, and measure latency and memory use on the intended hardware. Any unsupported-input or quality policy should be explicit and evaluated rather than silently treating failures as genuine.

With a sufficiently diverse labelled attack set, I would also revisit supervised or hybrid learning and synthetic augmentation. The purpose would be to establish whether these methods improve independent real-attack performance. Further optimisation on the current 23 attacks would provide less useful evidence than a stronger evaluation design.

Consider other approaches Finally, I would explore more realistic synthetic augmentation, including CycleGAN-based image translation to approximate screen-recapture appearance and a 3D rendering pipeline to simulate different displays, viewing angles, lighting conditions and reflections. Reproducing fine screen artefacts would also require modelling display pixel structure and camera sampling. With sufficient real data to develop and validate these methods, the generated images could support supervised or hybrid PAD models. Their value would be assessed by improvements on independently collected real attacks, rather than visual realism or performance on synthetic images alone.